

CO 250 Review 2: IP and NLP formulations

Integer program (IP) formulations

Restricts variables to take only integer values. Practical in many situations, but difficult to solve.

Two useful techniques:

- ① Binary variables: they take only 0 or 1 as values. Useful for indicating if some condition happens.

Example: We have n students $1, \dots, n$, to be divided into k teams $1, \dots, k$.

We need to decide "which student goes into which team?"

Define x_{ij} to represent $x_{ij} = \begin{cases} 1 & \text{student } i \text{ is in team } j \\ 0 & \text{otherwise} \end{cases}$, $i=1, \dots, n$ $j=1, \dots, k$.

For a student i , $x_{i1} + \dots + x_{ik} = 1$ (each student is in one team).

Binary constraints: $0 \leq x_{ij} \leq 1$, $x_{ij} \in \mathbb{Z}$. (Or, $x_{ij} \in \{0, 1\}$. Only use this notation for binary variables.)

- ② Logical conditions: "this or that" "this implies that" etc.

Example from lectures: Knapsack problem. Shipping crates.

- Each crate type has a **weight** and a **value**:

Type	1	2	3	4	5	6
weight (lbs)	30	20	30	90	30	70
value (\$)	60	70	40	70	20	90

- The **total weight** of crates shipped must not exceed 10,000 lbs.

Logical condition to implement: either at least 4 crates of type 1 or 2, or at least 4 crates of type 5 or 6.

Define a binary variable y to represent:

- If $y=1$, then $x_1 + x_2 \geq 4$
- If $y=0$, then $x_5 + x_6 \geq 4$.

Finding the constraints take some trial and error.

We want $x_1 + x_2 \geq 4$ when $y=1$, and $x_1 + x_2 \geq 0$ when $y=0$. $x_1 + x_2 \geq 4y$ works

We want $x_5 + x_6 \geq 4$ when $y=0$, and $x_5 + x_6 \geq 0$ when $y=1$. $x_5 + x_6 \geq 4(1-y)$ works
 $y \in \{0, 1\}$.

In this case, it is possible that both conditions hold. Exercise: How to implement exactly one condition holds?

Important concept

When we define variables, we give the intended interpretations for them. However, the computer only sees the objective function and the constraints, it does not know the meaning of your variables. So it is necessary to implement the meaning of the variables in the constraints.

Example: Suppose we can only send up to 3 types of crates.

Define variables z_i : $z_i = \begin{cases} 1 & \text{we send type } i \\ 0 & \text{otherwise.} \end{cases}$ $z_i \in \{0, 1\}$.

Additional constraint: $z_1 + \dots + z_6 \leq 3$.

This is not enough. We need to ensure that when $z_i=0$, we send 0 of type i $x_i=0$.

More constraints: $x_i \leq 10000z_i$. [when $z_i=0$, this forces $x_i=0$. When $z_i=1$, this gives $x_i \leq 10000$, which does not limit x_i since x_i can never exceed 10000]

Nonlinear program (NLP) formulation

General form : $\min f(x)$ $f, g_1, \dots, g_k$ are any functions.
s.t. $g_1(x) \leq 0$
 $\vdots$
 $g_k(x) \leq 0$

Formulates more situations, much more difficult to solve.

Tricks: If we want a variable x to be an integer, add the constraint $\sin(\pi x) = 0$.
For a binary variable x , add the constraint $x(x-1) = 0$.

Preview: When functions are "convex", it is easier to solve. Convex roughly means graph is facing upwards.

